

# README

Setup & run instructions

Multi-Agent Incident Response System — Track 2 Capstone  
[github.com/bgolsen/agents-final-project](https://github.com/bgolsen/agents-final-project)

# Multi-Agent Incident Response System

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Capstone project, Track 2 (Multi-Agent Incident Response System). See [ARCHITECTURE.md](#) for the full design writeup, diagrams, and technical analysis, and [SCRIPT.md](#) for the demo video script. This README is setup/run instructions only. PDF versions of all three ( [README.pdf](#) , [ARCHITECTURE.pdf](#) , [SCRIPT.pdf](#) ) are included at the repo root for easier offline reading; regenerate them after editing the `.md` sources with `python .pdfbuild/build_pdfs.py` (requires Chrome and `npm` for mermaid-cli — see the script for details).

A simulated production incident comes in as an alert. Four **Google ADK** agents, each its own **A2A** server, collaborate under a rule-based Coordinator to triage it, diagnose root cause against a knowledge base of past incidents, plan a risk-ranked remediation, stop for **human approval**, execute the approved plan, and write a post-mortem. **n8n** fronts the system for alert intake/routing; **LangSmith** traces the full reasoning chain.

Demo video: `<add your unlisted YouTube/Loom link here before submitting>`

## Prerequisites

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- Python 3.11+ (developed on 3.12)
- (Optional but recommended) a [Google AI Studio API key](#) for real Gemini-driven reasoning. **Without one, the system still runs end-to-end** — every agent has a deterministic rule-based fallback (see [ARCHITECTURE.md](#) §6) that fires automatically when the LLM call fails, which is exactly how this project was developed and tested.
- (Optional) a [LangSmith API key](#) to see traces. Without one, tracing is a documented no-op (see [ARCHITECTURE.md](#) §3).
- (Optional) Docker, to run n8n for the alerting/routing workflow via `docker compose up -d` (see below) rather than the CLI demo posting directly to the coordinator. Free/open-source, no n8n.io account needed.

# Setup

```
python -m venv .venv
# Windows:
.venv\Scripts\activate
# macOS/Linux:
source .venv/bin/activate

pip install -r requirements.txt      # or requirements-dev.txt to also run tests
cp .env.example .env                # then fill in GOOGLE_API_KEY / LANGSMITH_API_KEY
```

## Run the demo (single command)

```
python -m incident_response.run_incident
```

This starts the 4 specialist A2A servers (ports 8001-8004) and the coordinator API (port 8110) as subprocesses, lets you pick one of 3 canned incident scenarios, runs triage → diagnosis → planning, prints each agent's structured output, **stops and prompts you** to approve/reject/modify the remediation plan (the HITL checkpoint), then executes and prints the post-mortem. Processes are torn down automatically on exit. Each subprocess's own stdout/stderr (model calls, ADK's internal retry-on-validation-failure noise, etc.) is written to `logs/<agent>.log` instead of the terminal, so the demo output stays readable — check those files if something looks wrong.

Useful flags:

```
python -m incident_response.run_incident --scenario auth-cert-expiry # skip the menu
python -m incident_response.run_incident --auto-approve              # non-interactive
(scripted demo/CI)
python -m incident_response.run_incident --no-spawn                  # reuse already-
running servers
```

The three canned scenarios ( `incident_response/data/scenarios.py` ) each inject a different simulated failure so you can see error handling in action:

| scenario_id                | what's wrong                                  | what fails once (chaos injection)                            |
|----------------------------|-----------------------------------------------|--------------------------------------------------------------|
| checkout-latency-spike     | downstream payment-gateway timeouts cascading | the log API                                                  |
| auth-cert-expiry           | expired TLS certificate                       | the knowledge-base search                                    |
| payment-db-pool-exhaustion | leaked DB connections after a deploy          | the metrics API, and the first remediation-execution attempt |

## Live incident dashboard & report

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Terminal output only shows so much. While the system is running (via either `run_incident.py` or the n8n workflow below), open **`http://localhost:8110/dashboard`** in a browser — it lists **every incident the coordinator has ever handled**, not just the current one: on startup the coordinator reloads every `runs/<incident_id>.json` it has saved, so history survives restarts, with each one's status (`awaiting_approval`, `closed`, `rejected`, ...) color-coded and the list auto-refreshing while any incident is still in flight. Click into one for the full report at `/incidents/{id}/report`: each agent's actual structured output (triage anomalies, the diagnostic agent's sub-question decomposition and ranked hypotheses with cited evidence and confidence, the remediation goal/step plan with risk and rollback, the execution log, the postmortem) rendered as readable HTML, with a raw-JSON toggle at the bottom for the full record. This page updates in real time as each agent finishes — leave it open during a live demo and watch each section fill in — because the coordinator mutates the same in-memory `IncidentState` object that the page reads, so progress is visible mid-run, not just after the incident closes.

**The HITL checkpoint itself lives on this page, not just in a terminal.** When an incident reaches `awaiting_approval`, the report page shows an actual form — your name, notes, a dropdown of which goal to run if you modify — with Approve / Modify / Reject buttons that post straight to the coordinator and redirect back to the (now-updated) report. This is the only way to act on an incident that wasn't started from `run_incident.py`'s interactive prompt — e.g. anything filed through n8n or curl — since the dashboard has no terminal to prompt in. (The page intentionally stops auto-refreshing while a decision is pending, so a timed reload can't wipe out what you're mid-typing.)

## Run the pieces individually

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```
# one terminal per agent, or use scripts/start_agents.ps1 on Windows
python -m incident_response.a2a_server monitoring      # :8001
python -m incident_response.a2a_server diagnostic     # :8002
python -m incident_response.a2a_server remediation    # :8003
python -m incident_response.a2a_server postmortem     # :8004
python -m incident_response.api                       # :8110 (coordinator + HITL API)

# then drive it with curl, or:
python -m incident_response.run_incident --no-spawn
```

Each agent's A2A card is inspectable directly: `curl http://localhost:8001/.well-known/agent-card.json`

Coordinator API surface (also used by n8n):

| Endpoint                                                                                                             | Purpose                                                                                                        |
|----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| POST /incidents {"scenario_id": " ... "}                                                                             | runs triage→diagnosis→planning, returns state at awaiting_approval                                             |
| GET /incidents/{id}                                                                                                  | full incident state                                                                                            |
| GET /approvals/{id}                                                                                                  | lightweight poll target (used by n8n's Wait loop)                                                              |
| POST /incidents/{id}/decision {"decision": "approve\ reject\ modify", "reviewer": " ... ", "modified_goal": " ... "} | the HITL checkpoint (JSON; used by n8n/curl)                                                                   |
| POST /incidents/{id}/decision-form (HTML form fields)                                                                | the HITL checkpoint (used by the dashboard's Approve/Modify/Reject buttons; redirects back to the report page) |
| POST /notify                                                                                                         | mock notification sink (stands in for Slack/PagerDuty)                                                         |
| GET /dashboard                                                                                                       | human-readable live list of every incident, including past ones reloaded from runs/ on startup                 |
| GET /incidents/{id}/report                                                                                           | human-readable live view of one incident, including the HITL form when applicable                              |

## n8n workflow (alerting & routing)

n8n is free/open-source and fully self-hosted here — no n8n.io account needed. The owner email+password n8n asks for on first run is purely local to your own container (n8n's own user-management DB), not an external service.

**Start a dedicated n8n instance** (won't collide with any n8n you already have running elsewhere — it uses port 5679, not the default 5678):

```
docker compose up -d # starts n8n at http://localhost:5679
```

First time only: open <http://localhost:5679>, it'll prompt you to create that local owner account, then land you on an empty workflow list.

**Import the workflow:** the "..." menu (top right of the editor) → *Import from file...* → select `n8n/incident_response_workflow.json`. All 11 nodes and their connections should appear immediately. Click **Publish** (top right) so its production webhook is always listening — without publishing, n8n only listens for one call at a time in "test" mode.

**Verified working config:** the workflow's `Config` node already points `coordinator_base` at `http://host.docker.internal:8110` — required because n8n runs inside Docker, where `localhost` refers to the *container*, not your host machine where the coordinator API actually listens. (This was confirmed by actually running it: pointing it at `localhost:8110` fails with "the service refused the connection" on the `Start Incident` node — a good one to know about if you ever repoint it.)

### Trigger it:

```
curl -X POST http://localhost:5679/webhook/incident-alert \
-H "Content-Type: application/json" -d '{"scenario_id": "checkout-latency-spike"}
```

This returns immediately once routing/notification is done (e.g.

`{"incident_id": "INC- ... ", "status": "awaiting_approval", "routed_to": "pager-oncall"}` ) while the workflow keeps running in the background, polling `/approvals/{id}` every 5s until a human decides — approve/reject via `POST /incidents/{id}/decision` as above, or the live dashboard.

**To see the incident data flowing through n8n itself** (not just the coordinator's own dashboard): open the **Executions** tab in n8n, click the run, then click any node in the execution graph — the side panel shows exactly what that node received as input and produced as output (e.g. the `Alert Webhook` node shows the raw HTTP headers/body of the alert that came in; `Compute Routing` shows the severity-based channel decision it computed). Every node in a finished execution is inspectable this way.

To route the CLI demo's alert through n8n instead of posting directly to the coordinator, set

```
N8N_ALERT_WEBHOOK_URL=http://localhost:5679/webhook/incident-alert in .env .
```

## LangSmith tracing

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With `LANGSMITH_API_KEY` set in `.env`, every incident produces one nested run tree in your `LANGSMITH_PROJECT` (default `incident-response-adk`) — expand it to see the triage/diagnosis/remediation/execution/post-mortem stages in order, tagged `incident:<id>`, including which calls ran in degraded (fallback) mode.

## Tests

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```
pip install -r requirements-dev.txt
pytest tests/ -v
```

Covers the simulated data layer (KB ranking, chaos injection), schema validation, the deterministic fallback reasoning for all 4 agents, and the coordinator's resilience/HITL-goal-selection logic — all runnable without any API key or live agent servers.

## Project layout

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|                                     |                                                                                                                |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------|
| incident_response/                  |                                                                                                                |
| agents/                             | monitoring / diagnostic / remediation / postmortem LlmAgents<br>(each with an in-agent deterministic fallback) |
| a2a_server.py                       | launches one agent as a standalone A2A (Starlette) server                                                      |
| coordinator.py                      | orchestration, HITL gate, retry/fallback, execution                                                            |
| api/server.py                       | FastAPI surface used by n8n and the CLI demo                                                                   |
| api/report.py                       | renders the live HTML dashboard/incident-report pages                                                          |
| data/                               | simulated telemetry, knowledge base, remediation catalog,<br>canned incident scenarios (with chaos injection)  |
| schemas.py                          | Pydantic contracts agents hand off between each other                                                          |
| tracing.py                          | LangSmith helpers                                                                                              |
| run_incident.py                     | CLI demo entrypoint                                                                                            |
| n8n/incident_response_workflow.json | importable alert intake/routing workflow                                                                       |
| docker-compose.yml                  | dedicated n8n instance (port 5679)                                                                             |
| tests/                              | pytest suite (no API key required)                                                                             |
| ARCHITECTURE.md                     | full design writeup + diagrams                                                                                 |

## Known limitations

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See `ARCHITECTURE.md` §9 for the full discussion (fallback ranking quality, per-process chaos budget, in-memory incident store, no auth on approvals).